

Ventilation Sizing Summary for HP-101-FITNESS

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

05/29/2026
1:23 PM

1. Summary

Ventilation Sizing Method **Sum of Space OA Airflows**
Design Ventilation Airflow Rate **318** CFM
Design Ventilation Airflow Rate, Corrected for Exhaust Air **335** CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
HP-101-FITNESS										
141-FITNESS	1117.5	11.2	991.7	20.00	0.06	0.0	0.0	0.00	290.6	335.3
142-GUEST LAUNDRY	162.0	1.6	223.3	5.00	0.12	0.0	0.0	0.00	27.5	0.0
Totals			1215.0						318.1	

Ventilation Sizing Summary for HP-102-OFFICES

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

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1. Summary

Ventilation Sizing Method Sum of Space OA Airflows
Design Ventilation Airflow Rate 73 CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
HP-102-OFFICES										
119-SALES OFFICE	109.9	1.0	43.6	5.00	0.06	0.0	0.0	0.00	11.6	0.0
119-WORK AREA	407.7	3.3	142.1	5.00	0.06	0.0	0.0	0.00	40.8	0.0
122-GM OFFICE	120.3	1.0	58.2	5.00	0.06	0.0	0.0	0.00	12.2	0.0
123-CLOSET	25.2	0.0	5.0	5.00	0.06	0.0	0.0	0.00	1.5	0.0
124-RETAIL STORAGE	110.9	0.0	83.6	5.00	0.06	0.0	0.0	0.00	6.7	0.0
Totals			332.5						72.7	

Ventilation Sizing Summary for HP-103-BOH EMPLOYEE

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

05/29/2026
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1. Summary

Ventilation Sizing Method Sum of Space OA Airflows
Design Ventilation Airflow Rate 80 CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
HP-103-BOH EMPLOYEE										
125-BOH CORRIDOR	140.3	0.0	18.4	0.00	0.06	0.0	0.0	0.00	8.4	0.0
132-BREAK ROOM	389.1	9.7	556.4	5.00	0.06	0.0	0.0	0.00	72.0	0.0
133-STAFF RR	64.3	0.0	9.5	0.00	0.00	0.0	0.0	0.00	0.0	75.0
Totals			584.3						80.4	

Ventilation Sizing Summary for HP-104-BOH LAUNDRY

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Project: HAP_HamptonInn(Hilton)
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1. Summary

Ventilation Sizing Method Sum of Space OA Airflows
Design Ventilation Airflow Rate 188 CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
HP-104-BOH LAUNDRY										
126-BOH LAUNDRY	871.6	8.7	830.8	5.00	0.12	0.0	0.0	0.00	148.2	0.0
127 - DRYER ENCLOSURE	117.8	1.2	318.0	5.00	0.12	0.0	0.0	0.00	20.0	0.0
128-LINEN STORAGE	171.4	0.0	118.7	5.00	0.06	0.0	0.0	0.00	10.3	0.0
129-STORAGE	48.6	0.0	8.7	5.00	0.06	0.0	0.0	0.00	2.9	0.0
130-LINEN CLOSET	108.1	0.0	16.5	5.00	0.06	0.0	0.0	0.00	6.5	0.0
Totals			1292.7						187.9	

Ventilation Sizing Summary for HP-105-LOBBY/MARKET

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Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

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1. Summary

Ventilation Sizing Method Sum of Space OA Airflows
Design Ventilation Airflow Rate 298 CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
HP-105-LOBBY/MARKET										
101/102-MARKET/RECEPTION	651.0	19.5	579.1	7.50	0.06	0.0	0.0	0.00	185.7	0.0
103-LOBBY (entry)	650.9	9.8	338.6	7.50	0.06	0.0	0.0	0.00	112.2	0.0
Totals			917.7						297.9	

Ventilation Sizing Summary for HP-106-DINING LARGE

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
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1. Summary

Ventilation Sizing Method Sum of Space OA Airflows
Design Ventilation Airflow Rate 524 CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
HP-106-DINING LARGE										
105A DINING	1092.2	43.7	832.6	7.50	0.18	0.0	0.0	0.00	524.3	0.0
Totals			832.6						524.3	

Ventilation Sizing Summary for HP-107-BUFFET/SERVING

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
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05/29/2026
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1. Summary

Ventilation Sizing Method **Sum of Space OA Airflows**
Design Ventilation Airflow Rate **164** CFM
Design Ventilation Airflow Rate, Corrected for Exhaust Air **224** CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
HP-107-BUFFET/SERVING										
106-BREAKFAST/BUFFETT	747.0	10.0	779.6	7.50	0.12	0.0	0.0	0.00	164.3	224.1
Totals			779.6						164.3	

Ventilation Sizing Summary for HP-108-FOOD PREP

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
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05/29/2026
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1. Summary

Ventilation Sizing Method **Sum of Space OA Airflows**
Design Ventilation Airflow Rate **188** CFM
Design Ventilation Airflow Rate, Corrected for Exhaust Air **487** CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
HP-108-FOOD PREP										
107-FOOD PREP	695.6	13.9	1080.0	7.50	0.12	0.0	0.0	0.00	187.8	486.9
Totals			1080.0						187.8	

Ventilation Sizing Summary for HP-109-DINING FLEX

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

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1. Summary

Ventilation Sizing Method Sum of Space OA Airflows
Design Ventilation Airflow Rate 174 CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
HP-109-DINING FLEX										
109-DINING/FLEX	362.7	14.5	361.4	7.50	0.18	0.0	0.0	0.00	174.1	0.0
Totals			361.4						174.1	

Ventilation Sizing Summary for HP-110-DINING SMALL

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
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1. Summary

Ventilation Sizing Method Sum of Space OA Airflows
Design Ventilation Airflow Rate **342** CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
HP-110-DINING SMALL										
105B - DINING	713.0	28.5	702.8	7.50	0.18	0.0	0.0	0.00	342.2	0.0
Totals			702.8						342.2	

Ventilation Sizing Summary for HP-111-MEETING

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

05/29/2026
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1. Summary

Ventilation Sizing Method Sum of Space OA Airflows
Design Ventilation Airflow Rate 180 CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
HP-111-MEETING										
111-MEETING ROOM	658.9	26.4	1142.6	5.00	0.06	0.0	0.0	0.00	171.3	0.0
111-MEETING STORAGE	149.9	0.0	138.3	5.00	0.06	0.0	0.0	0.00	9.0	0.0
Totals			1280.8						180.3	

Ventilation Sizing Summary for HP-112-ENGINEERING

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

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1. Summary

Ventilation Sizing Method Sum of Space OA Airflows
Design Ventilation Airflow Rate 23 CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
HP-112-ENGINEERING										
134-ENGINEERING	232.6	1.9	77.4	5.00	0.06	0.0	0.0	0.00	23.3	0.0
Totals			77.4						23.3	

Ventilation Sizing Summary for HP-STAIRS1

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

05/29/2026
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1. Summary

Ventilation Sizing Method Sum of Space OA Airflows
Design Ventilation Airflow Rate 0 CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
HP-STAIRS1										
109-STAIRS 1	196.0	0.0	139.0	0.00	0.00	0.0	0.0	0.00	0.0	0.0
200-STAIRS 1	196.0	0.0	90.4	0.00	0.00	0.0	0.0	0.00	0.0	0.0
300-STAIRS 1	201.7	0.0	89.7	0.00	0.00	0.0	0.0	0.00	0.0	0.0
400-STAIRS 1	203.1	0.0	90.1	0.00	0.00	0.0	0.0	0.00	0.0	0.0
500-STAIRS 1	196.0	0.0	102.4	0.00	0.00	0.0	0.0	0.00	0.0	0.0
Totals			511.6						0.0	

Ventilation Sizing Summary for HP-STAIRS2

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

05/29/2026
1:23 PM

1. Summary

Ventilation Sizing Method Sum of Space OA Airflows
Design Ventilation Airflow Rate 0 CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
HP-STAIRS2										
143-STAIRS 2	201.3	0.0	141.0	0.00	0.00	0.0	0.0	0.00	0.0	0.0
210-STAIRS 2	202.7	0.0	112.3	0.00	0.00	0.0	0.0	0.00	0.0	0.0
310-STAIRS 2	198.6	0.0	111.8	0.00	0.00	0.0	0.0	0.00	0.0	0.0
410-STAIRS 2	199.1	0.0	109.7	0.00	0.00	0.0	0.0	0.00	0.0	0.0
510 STAIRS 2	199.1	0.0	132.1	0.00	0.00	0.0	0.0	0.00	0.0	0.0
Totals			606.9						0.0	